Leaf and Non-Leaf Node Counting in a General Tree

Time Complexity

• Tree Construction: O(n).

• Counting Leaf and Non-Leaf Nodes (Preorder Traversal): O(n).

• Displaying Leaf and Non-Leaf Nodes: O(n).

Space Complexity

• Tree Storage: O(n).

• Recursive Traversal: O(h), where h is the height of the tree.

• Leaf and Non-Leaf Lists: O(n).

• Overall Auxiliary Space: O(n).

Here, n is the number of nodes and h is the height of the tree.